

ViSurf: Visual Supervised-and-Reinforcement Fine-Tuning for Large Vision-and-Language Models

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Abstract

Post-training Large Vision-and-Language Models (LVLMs) typically involves Supervised Fine-Tuning (SFT) for knowledge injection or Reinforcement Learning with Verifiable Rewards (RLVR) for performance enhancement. However, SFT often leads to sub-optimal performance, while RLVR remains constrained by the model’s internal knowledge base. While a sequential SFT → RLVR pipeline can be used, it introduces significant computational overhead and suffers from catastrophic forgetting. To address these limitations, we propose ViSurf (**Visual Supervised-and-Reinforcement Fine-Tuning**), a unified, single-stage paradigm that integrates the strengths of both SFT and RLVR. By analyzing their training objectives, we establish a unified framework that injects ground-truth labels directly into RLVR rollouts, facilitating simultaneous external supervision and internal reinforcement. Furthermore, we introduce three novel reward control strategies to ensure training stability and optimization. Extensive experiments demonstrate that ViSurf consistently outperforms standalone SFT, RLVR, and the traditional two-stage pipeline across diverse benchmarks. In-depth analysis corroborates these findings, validating the derivation and design principles of ViSurf.

1. Introduction

Developing Large Vision-and-Language Models (LVLMs) capable of excelling across diverse visual perception tasks represents a pivotal frontier in visual intelligence. To date, research has predominantly relied on two distinct post-

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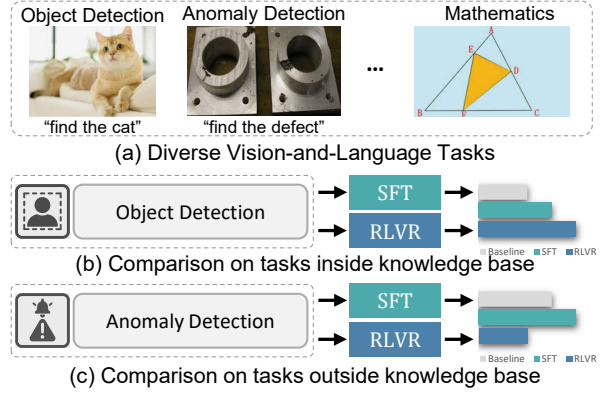


Figure 1. (a) Examples of vision-and-language tasks. (b) For tasks within LVLMs’ knowledge base, RLVR performs better than SFT. (c) For tasks that exceed LVLMs’ knowledge, SFT performs better, whereas RLVR performs worse than baseline.

training paradigms: Supervised Fine-Tuning (SFT) (Liu et al., 2023b; Wang et al., 2024c; Bai et al., 2025) and Reinforcement Learning with Verifiable Rewards (RLVR) (Liu et al., 2025b;c).

However, these paradigms exhibit a fundamental dichotomy in their strengths and weaknesses. SFT directly optimizes models using expert-annotated data, providing explicit external guidance that enables the model to approximate target distributions. Despite this, SFT often yields sub-optimal generalization and is prone to catastrophic forgetting of pre-trained knowledge. Conversely, RLVR, particularly on-policy methods like Group Relative Policy Optimization (GRPO) (Shao et al., 2024) and Dynamic Sampling Policy Optimization (DAPO) (Yu et al., 2025), leverages internal reinforcement signals to optimize policies based on rollout evaluations. While RLVR effectively mitigates forgetting and enhances performance, its efficacy is fundamentally capped by the model’s inherent knowledge boundaries; performance typically stagnates when tasks extend beyond the initial policy’s distribution. Our evaluation across diverse vision-language benchmarks, summarized in Figure 1, confirms this phenomenon: SFT excels in out-of-distribution knowledge acquisition, whereas RLVR thrives on tasks aligning with pre-existing capabilities (see Section 3.2 for a detailed case study). While a sequential SFT → RLVR

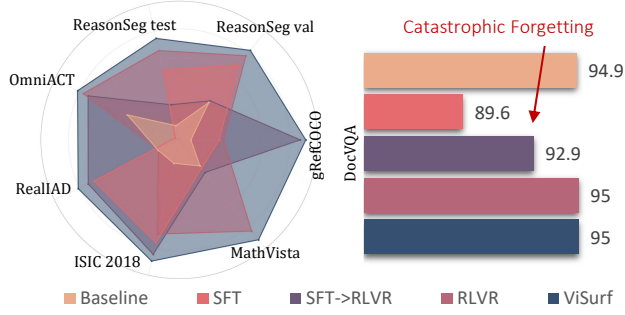


Figure 2. Radar Chart: ViSurf achieves superior performance across different training paradigms. Bar Chart: SFT and two-stage SFT $\rightarrow$ RLVR exhibit catastrophic forgetting.

pipeline attempts to marry these strengths, it introduces prohibitive computational overhead and remains vulnerable to forgetting during the initial SFT phase.

To overcome these challenges, we propose ViSurf (**V**isual **S**upervised-and-**R**einforcement **F**ine-Tuning), a unified, single-stage paradigm that integrates the complementary advantages of SFT and RLVR. We provide a rigorous analysis of the underlying objectives and gradients for both methods, theoretically demonstrating that their shared gradient patterns allow for integration into a singular ViSurf objective. Unlike existing approaches (Zhang et al., 2025; Ma et al., 2025) that simply sum the SFT and RLVR losses, ViSurf offers a theoretically grounded, unified perspective. While subtle differences exist, the gradient of ViSurf objective can be interpreted as a composite of the gradients from both SFT and RLVR. To ensure training stability within this unified framework, we introduce three novel reward control strategies for ground-truth labels: (i) preference alignment with policy rollouts, (ii) exclusion of "thinking" rewards for static labels, and (iii) reward smoothing to prevent optimization spikes. Consequently, the implementation of ViSurf is streamlined into an elegant process: interleaving ground-truth demonstrations with on-policy rollouts within a single, unified training phase.

We evaluate ViSurf across diverse domains, with a comparative summary presented in Figure 2. The results demonstrate that ViSurf consistently outperforms SFT, RLVR, and the SFT $\rightarrow$ RLVR pipeline. Notably, our method mitigates catastrophic forgetting, as evidenced by its stable performance on VQA benchmarks. Furthermore, our ablation study confirms the critical role of the proposed reward control mechanism, while in-depth analysis provides empirical support for our theoretical framework and offers insights into the operational principles of ViSurf. Our contributions are summarized as follows:

- **Unified Framework:** We introduce ViSurf, a theoretically grounded, single-stage paradigm that unifies SFT and RLVR to achieve simultaneous knowledge

injection and internal reinforcement.

- **Stability Optimization:** We design three novel reward control strategies to stabilize joint optimization, effectively balancing ground-truth guidance with on-policy exploration.
- **State-of-the-Art Performance:** ViSurf consistently outperforms standalone SFT, RLVR, and two-stage pipelines across diverse benchmarks, supported by in-depth analysis of its synergistic mechanics.

2. Related Works

2.1. Supervised Fine-tuning for LVLMs

Supervised Fine-Tuning (SFT) has emerged as the predominant paradigm for advancing Large Vision-Language Models (LVLMs), typically involving the refinement of pre-trained models using expert-annotated datasets. Prominent examples include the LLaVA-series (Liu et al., 2023b; Li et al., 2024a; Liu et al., 2024; An et al., 2025), QwenVL-series (Wang et al., 2024c; Bai et al., 2025), MGM-series (Li et al., 2024b; Zhong et al., 2024; Wang et al., 2025), Eagle (Chen et al., 2025) and InternVL-series (Chen et al., 2024b; Zhu et al., 2025), all of which leverage the SFT methodology. Beyond foundational training, SFT has demonstrated remarkable efficacy in tailoring LVLMs to specialized downstream tasks, such as image quality assessment (You et al., 2025) and autonomous driving (Xu et al., 2025).

2.2. Reinforcement Learning for LVLMs

Beyond supervised methods, Reinforcement Learning (RL) has become a cornerstone for aligning Large Vision-Language Models (LVLMs) with human expectations. Established techniques like DPO (Rafailov et al., 2023) necessitate extensive human-labeled preference datasets, which are often prohibitively expensive to curate. Similarly, PPO (Schulman et al., 2017) relies on a robust reward model to supervise the policy’s outputs. To mitigate these constraints, Reinforcement Learning from Verifiable Rewards (RLVR) algorithms, such as GRPO (Shao et al., 2024) and DAPO (Yu et al., 2025), leverage verification functions to evaluate model performance. The potency of this paradigm for LVLMs is underscored by recent advancements (Liu et al., 2025b;d;c;a; Huang et al., 2025), most notably in SegZero (Liu et al., 2025b) and VisualRFT (Liu et al., 2025d).

3. ViSurf

We begin by analyzing the objective functions of SFT and RLVR in Section 3.1. A case study in Section 3.2 then highlights the limitations of both methods. To address these limitations, we introduce ViSurf, detailing its design and its theoretical alignment with SFT and RLVR in Section 3.3.

Subsequently, Section 3.4 presents three novel mechanisms for reward control during training, and Section 3.5 a rigorous analysis of the underlying optimization dynamics.

3.1. Preliminary

Let π_θ denote a large vision-and-language model (LVLM), parameterized by θ . Common post-training paradigms for optimizing π_θ include Supervised Fine-Tuning (SFT) and Reinforcement Learning with Verifiable Rewards (RLVR). Both SFT and RLVR utilize the same input dataset, $\mathcal{D}_{\text{input}} = \{(v_i, t_i)\}_{i=1}^N$, where v_i is a visual input, t_i is a textual input, and N is the dataset size.

Supervised Fine-Tuning (SFT) optimizes π_θ against a set of ground-truth labels, $\mathcal{D}_{\text{label}} = \{y_i\}_{i=1}^N$. The objective is to minimize the negative log-likelihood of the labels:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(v,t) \sim \mathcal{D}_{\text{input}}, y \sim \mathcal{D}_{\text{label}}} [\log \pi_\theta(y | v, t)], \quad (1)$$

where y corresponds to (v, t) . A more precise notation would be $(v, t, y) \sim \text{zip}(\mathcal{D}_{\text{input}}, \mathcal{D}_{\text{label}})$. Nevertheless, we retain the current notation, $(v, t) \sim \mathcal{D}_{\text{input}}, y \sim \mathcal{D}_{\text{label}}$, for clarity and ease of comparison in the subsequent discussion.

Reinforcement Learning with Verifiable Rewards (RLVR). We illustrate RLVR using the on-policy Group Relative Policy Optimization (GRPO) algorithm (Shao et al., 2024). GRPO optimizes the policy π_θ using a verifiable reward function, which typically combines measures of output format and accuracy (Guo et al., 2025; Liu et al., 2025c;b). For a given input $(v_i, t_i) \in \mathcal{D}_{\text{input}}$, the old policy $\pi_{\theta_{\text{old}}}$ (from a previous optimization step) generates a group of G rollouts $\{o_j\}_{j=1}^G$ by sampling with different random seeds. Each rollout o_j is then evaluated by a reward function $r(\cdot)$, resulting in a set of rewards $\{r(o_j)\}_{j=1}^G$. The advantage for each rollout is subsequently computed as follows:

$$\hat{A}_j = \frac{r(o_j) - \text{mean}(\{r(o_j)\}_{j=1}^G)}{\text{std}(\{r(o_j)\}_{j=1}^G)}, \quad (2)$$

The objective of RLVR is to minimize the equation:

$$\mathcal{L}_{\text{RLVR}}(\theta) = -\mathbb{E}_{(v,t) \sim \mathcal{D}_{\text{input}}, \{o_j\}_{j=1}^G \sim \pi_{\theta_{\text{old}}}} \left[\frac{1}{G} \sum_{j=1}^G \min \left\{ \frac{\pi_\theta(o_j | v, t)}{\pi_{\theta_{\text{old}}}(o_j | v, t)} \hat{A}_j, \right. \right. \\ \left. \left. \text{clip} \left(\frac{\pi_\theta(o_j | v, t)}{\pi_{\theta_{\text{old}}}(o_j | v, t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_j \right\} \right], \quad (3)$$

where the ϵ is a constant that controls the clipping boundary. For simplicity, both in equation and in our practical implementation, we remove the KL divergence term.

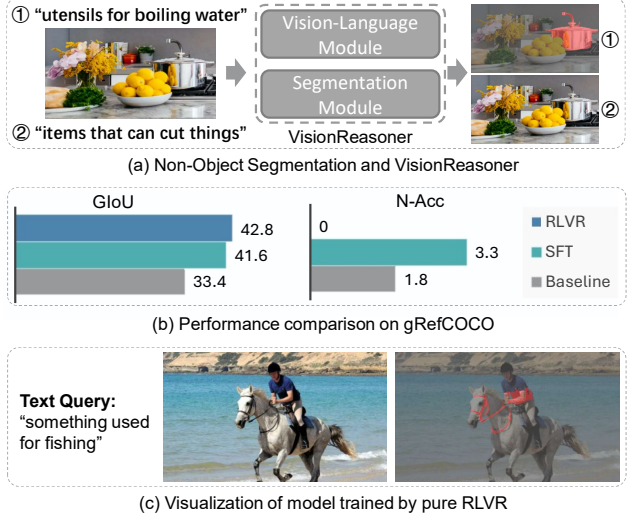


Figure 3. (a) Illustration on Non-Object Segmentation and VisionReasoner. (b) Performance comparison of SFT and RLVR on gRefCOCO. While RLVR achieves higher overall GIoU, it fails on non-object instructions. (c) Specifically, the RLVR model consistently outputs a mask even when no relevant object is detected.

3.2. Case Study: Non-Object Scenarios

We conduct a case study on non-object referring expression segmentation, a challenging task where instructions include both valid expressions and “incorrect” ones referring to non-existent objects. For this study, we utilize the VisionReasoner architecture (Liu et al., 2025c), which is initialized with Qwen2.5-VL (Bai et al., 2025) and SAM2 (Ravi et al., 2024) and has demonstrated strong performance on standard referring segmentation tasks. Figure 3(a) illustrates the non-object segmentation and the architecture of VisionReasoner. Experimental configurations are detailed in Section 4.1.

The gRefCOCO (Liu et al., 2023a) serves as the benchmark for this scenario. We compare gIoU (average IoU across images) and N-Acc (accuracy in identifying non-existent object). Quantitative and qualitative results are presented in Figure 3(b)-(c). Our analysis reveals that model trained by SFT achieves suboptimal gIoU performance but tends to learn to correctly identify the absence of objects. In contrast, model trained with RLVR attain higher overall gIoU scores, yet they often generate object masks even when no relevant object is present. This limitation arises because the RLVR model, relying solely on self-rollouts, lacks the corrective mechanism necessary to produce a correct “no object” output. In essence, RLVR drives performance gains through self-exploration, whereas SFT provides the external grounding necessary when exploration fails. This dichotomy motivates our central research question: *how can we efficiently synthesize the strengths of both training paradigms within a unified training stage?*

3.3. Combining SFT and RLVR

To harness the complementary benefits of SFT and RLVR, we propose ViSurf: a unified, single-stage post-training algorithm. This section elaborates on the derivation of the ViSurf algorithm and its relation to SFT and RLVR.

Gradient Analysis of SFT and RLVR. The gradient of SFT can be derived from Equation (1) as:

$$\nabla_{\theta} \mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(v,t) \sim \mathcal{D}_{\text{input}}, y \sim \mathcal{D}_{\text{label}}} [\nabla_{\theta} \log \pi_{\theta}(y | v, t)]. \quad (4)$$

In our practical implementation, we adopt a single-sampling, single-updating paradigm, which effectively bypasses the need for clipping operations. Therefore, the gradient of RLVR can be derived from Equation (3) using approximation $\pi_{\theta} \approx \pi_{\theta_{\text{old}}}$ and log-derivative trick:

$$\nabla_{\theta} \mathcal{L}_{\text{RLVR}}(\theta) = -\mathbb{E}_{(v,t) \sim \mathcal{D}_{\text{input}}, \{o_j\}_{j=1}^G \sim \pi_{\theta_{\text{old}}}} \left[\frac{1}{G} \sum_{j=1}^G \hat{A}_j \nabla_{\theta} \log \pi_{\theta}(o_j | v, t) \right]_{\theta \approx \theta_{\text{old}}}. \quad (5)$$

We observe that the gradients of the SFT and RLVR losses, $\nabla_{\theta} \mathcal{L}_{\text{SFT}}(\theta)$ and $\nabla_{\theta} \mathcal{L}_{\text{RLVR}}(\theta)$, share a similar form. The difference between them is the guidance signal (y vs. $\{o_j\}_{j=1}^G$) and coefficient (1 vs. $\hat{A}_j$).

Objective of ViSurf. To combine SFT and RLVR into a single stage, we design an objective function that naturally yields a gradient combining both $\nabla_{\theta} \mathcal{L}_{\text{SFT}}(\theta)$ and $\nabla_{\theta} \mathcal{L}_{\text{RLVR}}(\theta)$. Our key insight is to include the ground-truth label y as a high-reward sample within the RLVR framework. We construct an augmented rollout set $y \cup \{o_j\}_{j=1}^G$. Then the corresponding rewards are $r(y) \cup \{r(o_j)\}_{j=1}^G$. This formulation modifies the advantage calculation of rollouts in Equation (2) as follows:

$$\hat{A}_j = \frac{r(o_j) - \text{mean}(r(y) \cup \{r(o_j)\}_{j=1}^G)}{\text{std}(\{r(y) \cup \{r(o_j)\}_{j=1}^G\})}, \quad (6)$$

and the advantage of ground-truth y is calculated as:

$$\hat{A}_y = \frac{r(y) - \text{mean}(r(y) \cup \{r(o_j)\}_{j=1}^G)}{\text{std}(\{r(y) \cup \{r(o_j)\}_{j=1}^G\})}. \quad (7)$$

The objective of ViSurf is to minimize the equation:

Algorithm 1: ViSurf Optimization Step

Input:

policy model π_{θ} ; reward function $r(\cdot)$;
input data $\mathcal{D}_{\text{input}}$; label data $\mathcal{D}_{\text{label}}$

for $step = 1, \dots, M$ **do**

 Sample a mini-batch $\mathcal{B}_{\text{input}}$ and corresponding $\mathcal{B}_{\text{label}}$

 Update the old policy model $\pi_{\theta_{\text{old}}} \leftarrow \pi_{\theta}$

 Sample G outputs $\{o_j\}_{j=1}^G \sim \pi_{\theta_{\text{old}}}$

 for each $(v, t) \in \mathcal{B}_{\text{input}}$

 Compute rewards $\{r(o_j)\}_{j=1}^G$ for each sampled output o_j

 Compute rewards $r(y)$ for label $y \in \mathcal{B}_{\text{label}}$

 Compute $\hat{A}_j$ and $\hat{A}_y$ through relative advantage estimation

 Update the policy model π_{θ} using Equation (8)

Output: π_{θ}

$$\begin{aligned} \mathcal{L}_{\text{ViSurf}}(\theta) = & -\mathbb{E}_{(v,t) \sim \mathcal{D}_{\text{input}}, \{o_j\}_{j=1}^G \sim \pi_{\theta_{\text{old}}}, y \sim \mathcal{D}_{\text{label}}} \\ & \left[\frac{1}{G+1} \left(\sum_{j=1}^G \min \left\{ \frac{\pi_{\theta}(o_j | v, t)}{\pi_{\theta_{\text{old}}}(o_j | v, t)} \hat{A}_j, \right. \right. \right. \\ & \quad \left. \left. \left. \text{clip} \left(\frac{\pi_{\theta}(o_j | v, t)}{\pi_{\theta_{\text{old}}}(o_j | v, t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_j \right\} \right. \right. \\ & \quad \left. \left. + \min \left\{ \frac{\pi_{\theta}(y | v, t)}{\pi_{\theta_{\text{old}}}(y | v, t)} \hat{A}_y, \right. \right. \right. \\ & \quad \left. \left. \left. \text{clip} \left(\frac{\pi_{\theta}(y | v, t)}{\pi_{\theta_{\text{old}}}(y | v, t)}, 1 - \epsilon, 1 + \epsilon \right) \hat{A}_y \right\} \right) \right]_{\theta \approx \theta_{\text{old}}}. \quad (8) \end{aligned}$$

With the objective function demonstrated above, the pseudocode of ViSurf Optimization Step is shown in Algorithm 1.

Gradient Analysis of ViSurf. In our practical implementation, we adopt a single-sampling, single-updating paradigm, which effectively bypasses the need for clipping operations. Therefore, the gradient of Equation (8) can be derived using approximation $\pi_{\theta} \approx \pi_{\theta_{\text{old}}}$ and log-derivative trick:

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_{\text{ViSurf}}(\theta) = & -\mathbb{E}_{(v,t) \sim \mathcal{D}_{\text{input}}, \{o_j\}_{j=1}^G \sim \pi_{\theta_{\text{old}}}, y \sim \mathcal{D}_{\text{label}}} \\ & \left[\frac{1}{G+1} \left(\sum_{j=1}^G \hat{A}_j \nabla_{\theta} \log \pi_{\theta}(o_j | v, t) \right. \right. \\ & \quad \left. \left. + \hat{A}_y \nabla_{\theta} \log \pi_{\theta}(y | v, t) \right) \right]_{\theta \approx \theta_{\text{old}}}. \quad (9) \end{aligned}$$

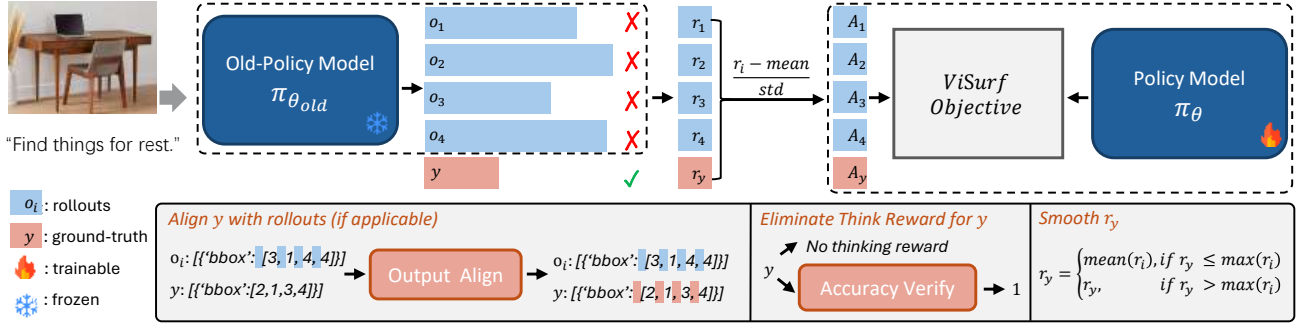


Figure 4. ViSurf Framework. Upper: The integration of external guidance y with internal guidance o_i , which is critical when self-rollouts are unsuccessful. Bottom: Three reward control strategies designed to regulate y , thereby preventing entropy collapse.

3.4. Reward Control for Ground-Truth Label

The advantage $\hat{A}_y$ for the ground-truth label y is always positive till now, as correct labels inherently receive higher rewards. However, this static setup is often sub-optimal; it can lead to reward hacking and suppresses the relative advantage $\hat{A}_j$ of self-generated rollouts, even when the policy has already produced correct trace and answers. Furthermore, maintaining the approximation $\pi_{\theta} \approx \pi_{\theta_{old}}$ requires ensuring that ground-truth data distributions remain aligned with self-generated rollouts. To mitigate these issues, we propose three reward control strategies:

Aligning Ground-truth Labels with Rollouts Preference. To ensure compatibility between the ground-truth annotations and the outputs generated by π_{θ} , we reformat the ground-truth data to match the model’s preferred stylistic patterns. For instance, we adjust the whitespace in JSON-like structures from $\{\text{'bbox':}[x1,y1,x2,y2]\}$ to $\{\text{'bbox':}[x1, y1, x2, y2]\}$ (i.e., adding space after the punctuation), because minor variations in punctuation can lead to significantly different tokenizations. This alignment minimizes the distribution shift between π_{θ} and $\pi_{\theta_{old}}$, thereby upholding the core assumptions $\pi_{\theta} \approx \pi_{\theta_{old}}$.

Eliminating Thinking Reward for Ground-truth Labels. Since the ground-truth labels lack annotated reasoning path, we assign a reasoning format score of zero to them when reasoning is needed. This ensures that the model learns to generate its own reasoning traces through self-rollouts without being biased or penalized by the absence of external reasoning annotations in the ground-truth data.

Smoothing the Reward for Ground-truth Labels. Prior to advantage estimation, we compare the maximum reward among generated rollouts, $\max\{r(o_j)\}_{j=1}^G$, against the ground-truth reward $r(y)$. If $\max\{r(o_j)\}_{j=1}^G \geq r(y)$, it indicates the policy model π_{θ} has already produced a high-quality output without external guidance. In this case, we set $r(y) = \text{mean}\{r(o_j)\}_{j=1}^G$. This smoothing ensures that the advantage for the ground-truth, $\hat{A}_y$, becomes zero (as per

Equation (7)), eliminating the external supervision signal.

3.5. Optimization Analysis During Training

To better analysis the optimization step, we reformulate the gradient Equation (9) as following:

$$\begin{aligned} \nabla_{\theta} \mathcal{L}_{\text{ViSurf}}(\theta) = & \underbrace{-\mathbb{E}_{\substack{(v,t) \sim \mathcal{D}_{\text{input}} \\ \{o_j\}_{j=1}^G \sim \pi_{\theta_{old}}}} \left[\frac{1}{G+1} \sum_{j=1}^G \hat{A}_j \nabla_{\theta} \log \pi_{\theta}(o_j | v, t) \right]}_{\text{RLVR Term}} \bigg|_{\theta \approx \theta_{old}} \\ & - \underbrace{\mathbb{E}_{\substack{(v,t) \sim \mathcal{D}_{\text{input}} \\ y \sim \mathcal{D}_{\text{label}}}} \left[\frac{1}{G+1} \hat{A}_y \nabla_{\theta} \log \pi_{\theta}(y | v, t) \right]}_{\text{SFT Term}} \bigg|_{\theta \approx \theta_{old}}. \end{aligned} \quad (10)$$

The RLVR term in Equation (10) is structurally identical to the standard RLVR gradient in Equation (5), differing only in its scaling coefficient ($\frac{1}{G+1} \hat{A}_j$ vs. $\frac{1}{G} \hat{A}_j$). Similarly, the SFT term in Equation (10) resembles the SFT gradient from Equation (4), with two key distinctions: (i) the coefficient is weighted by $\frac{1}{G+1} \hat{A}_y$ instead of 1, and (ii) the use of the approximation $\pi_{\theta} \approx \pi_{\theta_{old}}$. Crucially, Equation (9) integrates both the external guidance from SFT and the internal guidance from RLVR.

Building on the reward control strategy in Section 3.4, we analyze the dynamics of the terms in Equation (10) throughout training. As defined by Equations (6) and (7), the advantages $\hat{A}_j$ (for rollouts) and $\hat{A}_y$ (for the ground-truth) govern the balance between the RLVR and SFT terms. This balance is self-adaptive. When the policy fails to generate high-quality rollouts, $\hat{A}_j$ decreases (potentially becoming negative), while $\hat{A}_y$ remains high. Consequently, the SFT term dominates the policy update, providing strong external guidance from the ground-truth label. Conversely, when the policy successfully generates desirable rollouts, our reward control mechanism sets $\hat{A}_y \approx 0$, causing the optimization

Table 1. Comparison on different benchmarks in different domains under different training paradigms.

Method	Non-Object gRefCOCO val		Segmentation ReasonSeg val test		GUI OmniACT test	Anomaly ReallAD subset	Medical:Skin ISIC2018 test	Math MathVista test-mini	Avg
	gIoU	N-Acc	gIoU	gIoU	Acc	ROC_AUC	Bbox_Acc	Acc	
Baseline	33.4	1.8	56.9	52.1	60.4	50.1	78.8	68.2	50.2
SFT	41.6	3.3	63.8	60.3	55.4	65.5	91.7	68.3	56.2
RLVR	42.8	0.0	66.0	63.2	65.5	50.0	90.3	71.2	56.1
SFT → RLVR	65.0	52.1	57.2	55.2	64.5	66.9	93.6	68.5	65.4
ViSurf	66.6	57.1	66.5	65.0	65.6	69.3	94.7	71.6	69.6

to be dominated entirely by the RLVR term. This automatic shifting between learning modes is a core feature of the single-stage ViSurf paradigm.

Upper Bound Analysis. As mentioned above, our ViSurf is particularly beneficial when old policy model $\pi_{\theta_{old}}$ cannot generate correct rollouts. When the old policy model $\pi_{\theta_{old}}$ already achieves desirable rollouts, the SFT Term in Equation (10) equals to zero, thus the upper bound of Vi-Surf is the RLVR. However, when the policy model cannot generate desirable rollouts, the upper bound is better than using either SFT or RLVR alone.

4. Experiments

Section 4.1 details the experimental settings. We validate ViSurf across diverse domains in Section 4.2, followed by an ablation of the reward control design in Section 4.3. Finally, Section 4.4 provides a in-depth analysis of ViSurf.

4.1. Experimental Settings

We verify ViSurf on benchmarks across several domains, including Non-Object Segmentation (e.g., gRefCOCO (Liu et al., 2023a)), Reasoning Segmentation (e.g., ReasonSeg (Lai et al., 2024)), GUI Grounding (e.g., OmniACT (Kapoor et al., 2024)), Industrial Anomaly Detection (e.g., ReallAD (Wang et al., 2024a)), Medical Imaging (e.g., ISIC2018 (Codella et al., 2019)), and Mathematical Reasoning (e.g., MathVista (Lu et al., 2023)). Details of data split and evaluation protocols are provided in the Appendix C.

Implementation Details. We initialize ViSurf algorithm with Qwen2.5VL-7B (Bai et al., 2025) and adopt SAM2 (Ravi et al., 2024) if needed. We employ a constant learning rate of $1e-6$ for all methods, with a batch size of 32 for SFT and 16 for RLVR and ViSurf. We employ same training steps for fair comparison. For MathVista, the reward function consists of format and accuracy rewards. For other tasks, we adopt the rewards from VisionReasoner (Liu et al., 2025c), which include format accuracy, point accuracy, and bounding box accuracy rewards, etc.

Table 2. Comparison on VQA under different training paradigms.

Method	ChartQA	DocVQA_val
Baseline	83.8	94.9
SFT	80.8	89.6
RLVR	86.7	95.0
SFT → RLVR	85.0	92.9
ViSurf	87.4	95.0

4.2. Comparison of Different Training Paradigms

We compare different post-training paradigms and verify the effectiveness of ViSurf in various domains.

Main Results. A comparative analysis of various post-training paradigms is summarized in Table 1. Empirical evaluations demonstrate that ViSurf consistently surpasses existing methodologies across all evaluated domains, achieving a substantial average relative improvement of 38.6% over the baseline. This performance gain is most pronounced in challenging domains such as Non-Object and Anomaly, where the baseline initially struggles. This suggests that ViSurf is particularly effective at expanding a model’s capabilities in areas that exceed its inherent knowledge base. Conversely, in domains where the baseline already exhibits high proficiency, we observe more marginal incremental improvements. Our analysis also reveals that standard Supervised Fine-Tuning (SFT) leads to performance degradation on OmniACT, a phenomenon likely stemming from overfitting during the baseline’s original pre-training phase. In contrast, both RLVR and ViSurf successfully preserve the baseline’s foundational performance. Notably, in ReallAD and gRefCOCO (non-object detection), the pure RLVR approach actually underperforms the original model. We attribute this to the high frequency of incorrect self-generated rollouts, which introduce noise and hinder effective model optimization, a bottleneck that ViSurf effectively mitigates.

Catastrophic Forgetting. We evaluate the performance of ChartQA (Masry et al., 2022) and DocVQA (Mathew et al., 2021) after fine-tuning without VQA data. As illustrated in Table 2, VQA performance exhibits notable variation across different training paradigms. Both RLVR and ViSurf

Table 3. Employ ViSurf on Qwen2VL-7B.

Method	RealIAD subset	ISIC2018 test
	ROC_AUC	Bbox_Acc
Baseline	60.0	51.8
SFT	56.7	94.2
RLVR	57.1	90.5
SFT → RLVR	67.5	94.6
ViSurf	76.0	95.4

Table 4. Ablation of Reward Control Strategy in Section 3.4. The first row is the non-trained baseline. ‘Align’: Aligning ground-truth labels with rollouts; ‘Eliminate’: Eliminating thinking format reward for ground-truth labels; ‘Smooth’: Smoothing accuracy reward for ground-truth labels; ‘-’: not applicable.

Align	Eliminate	Smooth	gRefCOCO		ReasonSeg	MathVista
			val		val	testmini
			gIoU N-Acc		gIoU	Acc
-	-	-	33.4	1.8	56.9	68.2
×	✓	✓	59.0	40.2	63.6	—
✓	×	✓	72.9	74.1	58.2	67.1
✓	✓	×	61.0	45.7	62.7	66.8
✓	✓	✓	66.6	57.1	66.5	71.6

demonstrate robustness against catastrophic forgetting. In contrast, SFT and SFT → RLVR suffer from performance degradation, which is attributable to catastrophic forgetting.

ViSurf on Other Models. We apply ViSurf to the Qwen2VL-7B (Wang et al., 2024c). As shown in Table 3, our method consistently outperforms its counterparts.

4.3. Ablation of Reward Control

Table 4 presents an ablation study of the reward control strategy for ground-truth labels, detailed in Section 3.4.

Aligning Ground-truth Labels with Rollouts Preference. The empirical results underscore the critical importance of this strategy, as its ablation leads to consistent performance degradation across multiple datasets. This observation provides strong empirical validation for the theoretical requirement of $\pi_\theta \approx \pi_{\theta_{old}}$ presented in Equation (10).

Eliminating Thinking Reward for Ground-truth Labels. The results indicate that the reasoning strategy is critical for tasks requiring complex inference, such as those in ReasonSeg and MathVista, as it encourages the model to generate a reasoning process prior to delivering the final answer. Conversely, for the gRefCOCO dataset, where queries are typically limited to simple class types (e.g., “human”) and basic references (e.g., “woman on the right”), omitting the reasoning trace yields superior performance. This suggests that the necessity of explicit reasoning is contingent upon the complexity of the underlying task.

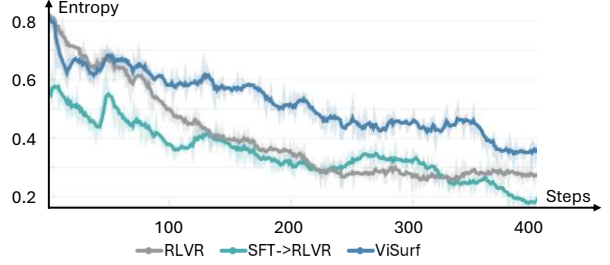


Figure 5. Entropy Analysis of RLVR, SFT → RLVR and ViSurf. ViSurf exhibits an initial drop, then converges slowly.

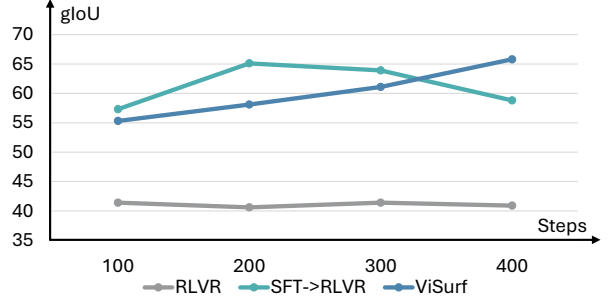


Figure 6. Performance on gRefCOCO in different training steps. ViSurf demonstrates greater stability as training proceeds.

Smoothing the Reward for Ground-truth Labels. The performance decline observed across datasets following the ablation of reward smoothing underscores its necessity. Concurrently, the results suggest that the SFT Term in Equation (10) becomes superfluous when the model’s self-rollouts already achieve a higher-quality solution.

4.4. In-depth Analysis

To facilitate a deeper understanding of ViSurf, we provide a detailed analysis of its behavior and properties.

Entropy Analysis During Training. Figure 5 compares the entropy of RLVR, SFT → RLVR and ViSurf. A higher entropy indicates greater exploratory behavior, while lower entropy signifies convergence toward certainty. We observe that ViSurf exhibits an initial entropy drop, indicating the model is fitting the external guidance. Subsequently, ViSurf converges at a slower rate than others, thereby effectively avoiding entropy collapse.

Training Stability. Figure 6 demonstrates that models trained with our method exhibit greater stability than those trained with pure RLVR and SFT → RLVR, as the performance of others decline with longer training. This observation confirms the effectiveness of our approach, indicating that the introduced external guidance acts as a constraint, which stabilizes the training process.

Boundary Analysis. As shown in Table 1, the performance gain of ViSurf is related to the baseline model’s performance.

Table 5. Comparison of different prompt design. ViSurf achieves satisfying results even without detailed formatting prompt.

Detailed Prompt		ReasonSeg	
		val (gIoU)	test (gIoU)
RLVR	✗	0.0	0.0
	✓	66.0	63.2
ViSurf	✗	62.3	57.8
	✓	66.4	65.0

Table 6. Comparison of training cost of different training paradigms with same batch size. Time for two-stage SFT → RLVR is estimated as the addition of SFT and RLVR.

Method	Mem / GPU (G) ↓	Time / Step (s) ↓
SFT	97.7	9.0
RLVR	81.8	22.7
SFT → RLVR	97.9	31.7
ViSurf	81.8	22.9

When the baseline performs poorly (e.g., below 50%), indicating its inadequacy for the task, our method yields a substantial improvement. Conversely, when the baseline already achieves high performance (e.g., above 50%), signifying a strong starting point, the upper bound of our method aligns with that of RLVR alone. This observation corroborates our theoretical analysis in Section 3.5.

Reduce the Burden of Prompt Design. The RLVR paradigm relies heavily on explicit instructions to guide the model toward generating rollouts in a specific format, such as *output with format 'point 2d': [2, 3]*. In contrast, ViSurf incorporates external guidance with desired format, thereby reducing the dependency on manual prompt engineering. Table 5 compares performance with and without detailed prompts (*prompts difference are shown in the supplementary materials*), demonstrating that our approach achieves consistent gains in both settings, whereas RLVR fails without explicit formatting instructions.

Training Cost. We conducted a comparative analysis of the per-step training for different fine-tuning paradigms. Each method is implemented using well-established frameworks: DeepSpeed (Microsoft & DeepSpeed Team, 2020) and TRL (Face, 2024) for SFT, and VeRL (Seed, 2024) for RLVR and ViSurf. The results indicate that while RLVR and ViSurf offer greater memory efficiency, they increase computational time, attributable to the overhead of generating rollouts.

4.5. Comparison with State-of-The-Arts

We compare ViSurf against state-of-the-art (SoTA) models on two visual perception tasks: gRefCOCO and ReasonSeg. We compare LISA (Lai et al., 2024), GSVA (Xia et al., 2024), SAM4MLLM (Chen et al., 2024a), SegZero (Liu et al., 2025b), VisionReasoner (Liu et al., 2025c). As shown

Table 7. Comparison with SoTAs. ‘-’ means not available.

Method	gRefCOCO		ReasonSeg	
	val		val	test
	gIoU	N-Acc	gIoU	gIoU
LISA-7B	61.6	54.7	53.6	48.7
GSVA-7B	66.5	62.4	-	-
SAM4MLLM-7B	69.0	63.0	46.7	-
Qwen2.5VL-7B + SAM2	41.6	3.3	56.9	52.1
SegZero-7B	-	-	62.6	57.5
VisionReasoner-7B	41.5	0.0	66.3	63.6
ViSurf (Qwen2.5VL-7B + SAM2)	72.9	74.1	66.5	65.0

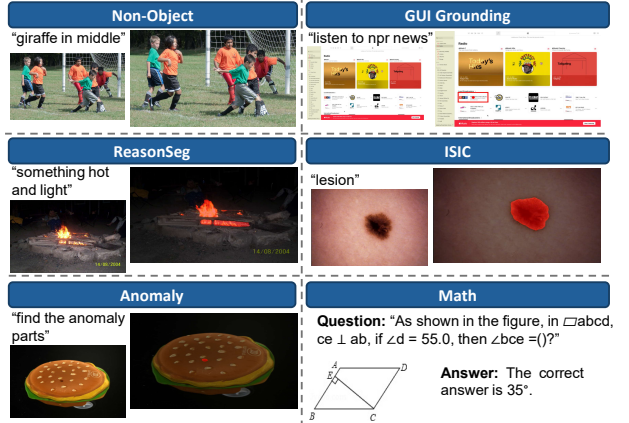


Figure 7. Visualization of ViSurf on various tasks.

in Table 7, ViSurf achieves the SoTA performance.

4.6. Qualitative Results

Qualitative results on various tasks are presented in Figure 7. The results demonstrate that models trained with ViSurf successfully solves multiple visual perception tasks.

5. Conclusion

We present ViSurf, a unified post-training paradigm designed to bridge the gap between Supervised Fine-Tuning (SFT) and Reinforcement Learning from Verifiable Rewards (RLVR). Driven by a rigorous theoretical analysis of their respective objectives and gradient dynamics, ViSurf integrates the strengths of both approaches into a single-stage framework. Our implementation strategically interleaves ground-truth labels with model-generated rollouts, utilizing three novel reward control strategies to maintain training stability. Empirical evaluations across diverse benchmarks demonstrate that ViSurf consistently outperforms standalone SFT, RLVR, and the traditional sequential SFT → RLVR pipeline. Our in-depth analysis provides deep insights into the framework’s behavior, effectively corroborating our theoretical foundations.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

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A. Limitations

The current reinforcement learning codebase relies on VeRL (Seed, 2024), which mainly support the Qwen-VL series (Bai et al., 2025; Wang et al., 2024c). Extending support to additional architectures is left for future work as framework compatibility improves.

B. Discussion and Future Work

The principal insight of ViSurf is the effective combination of RLVR’s internal reinforcement and the external guidance of SFT. Although the ground-truth labels in this work are limited to final answers, our ViSurf paradigm is inherently compatible to incorporate explicit reasoning traces. The flexibility also ensures compatibility with advanced techniques like knowledge distillation, where reasoning traces from larger models could be directly incorporated. We anticipate that this work will provide a foundation for future research in LVLMs’ post-training.

C. Detailed Experimental Settings

Non-Object Segmentation. The gRefCOCO (Liu et al., 2023a) includes queries that do not contain corresponding objects. The evaluation metrics are gIoU and N-Acc. We use Multi-objects-7K (Liu et al., 2025c) plus with 200 non-object data for training.

Reasoning Segmentation. The ReasonSeg (Lai et al., 2024) includes test samples that need reasoning for correct segmentation. It has 200 validation images and 779 test images. The evaluation metric is gIoU. We use Multi-objects-7K (Liu et al., 2025c) proposed in VisionReasoner (Liu et al., 2025c) for training.

GUI Grounding. The OmniACT (Kapoor et al., 2024) is a GUI grounding task for Desktop and Web. We randomly collect 6,101 samples in training split and verify on the test split. The accuracy is calculated as whether the predict point correctly locates inside the interest region.

Anomaly Detection. The RealIAD (Wang et al., 2024a) includes real-world, multi-view industrial anomaly. We derive 3,292 training samples and 2,736 test samples, ensuring the two sets are disjoint. We calculated the ROC_AUC.

Medical Image: Skin. The task one of ISIC2018 (Codella et al., 2019; International Skin Imaging Collaboration (ISIC), 2018) is lesion segmentation. It includes 2,594 training samples and 1,000 test samples. We measure the bbox_acc metric, which computes the ratio of predicted bboxes whose IoU with the ground truth exceeds 0.5.

MathVista. The MathVista-testmini (Lu et al., 2023) includes 1,000 diverse mathematical and visual tasks. We gather around 10k training data from WeMath (Qiao et al.,

2024), MathVision (Wang et al., 2024b), Polymath (Gupta et al., 2024), SceMQA (Liang et al., 2024), Geometry3K (Lu et al., 2021).

D. Illustration of Prompt Design

Prompt 2 shows prompt with detailed format instruction, where we provide desired output format for model. Prompt 1 shows prompt without detailed format instruction, where we simply write ‘answer here’ between answer tags.

Prompt 1: without detailed format instruction

```
“Please find ‘{{Question}}’ with bboxes and points.”
“Compare the difference between object(s) and find the most closely matched object(s).”
“Output the thinking process in <think> </think> and final answer in <answer> </answer> tags.”
“Output the bbox(es) and point(s) inside the interested object(s) in JSON format.”

i.e. <think>
thinking process here
</think>
<answer>
answer here
</answer>
```

Prompt 2: with detailed format instruction

```
“Please find ‘{{Question}}’ with bboxes and points.”
“Compare the difference between object(s) and find the most closely matched object(s).”
“Output the thinking process in <think> </think> and final answer in <answer> </answer> tags.”
“Output the bbox(es) and point(s) inside the interested object(s) in JSON format.”

i.e. <think>
thinking process here
</think>
<answer>
[{"bbox_2d": [10, 100, 200, 210],
"point_2d": [30, 110]},
{"bbox_2d": [225, 296, 706, 786],
"point_2d": [302, 410]}]
</answer>
```

E. Additional Explanation on pure RLVR

The performance of pure RLVR is heavily influenced by randomness. In rare cases, the old policy model $\pi_{\theta_{old}}$ can generate correct non-object outputs in the initial steps, thereby

achieving competitive performance (see Table 8), yet still marginally inferior to ViSurf.

Table 8. Comparison under different training paradigms.

Method	gRefCOCO val	
	gIoU	N-Acc
Baseline	33.4	1.8
SFT	41.6	3.3
RLVR	42.8	0.0
RLVR (rare)	62.9	49.3
SFT $\rightarrow$ RLVR	58.6	38.1
ViSurf	66.6	57.1

F. Non Object Training Data

The training data for gRefCOCO was adapted from the VisionReasoner training data (Liu et al., 2025c), from which we utilized 7k referring expression samples. To enhance the model’s ability to handle cases where no target object is present, we augmented this dataset with 200 non-object examples. These negative samples were generated by providing a question that is unanswerable given the image content and training the model to output an empty list (`<answer>[]</answer>`).

G. Data Mixture in Non-object Scenarios

We conduct a data sensitivity analysis for the non-object setting. As shown below, ViSurf’s performance consistently improves as the amount of non-object data increases. Notably, it can already detect absent objects with as few as 50 negative samples, demonstrating strong robustness even under limited annotated data.

Table 9. Performance comparison with different training data mixture on gRefCOCO.

Method	Obj	Non-Obj	gIoU	N-ACC
Baseline	–	–	33.4	1.8
ViSurf	7k	50	61.5	44.4
ViSurf	7k	200	66.6	57.1
ViSurf	6.7k	500	74.9	80.2